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Key Points:

- S-receiver functions reveal a sharp lithosphere–asthenosphere boundary at ~70 km depth beneath the southern Korean Peninsula
- Partial melt likely contributes to the sharp boundary and observed shear-wave velocity reduction
- Regional variations in lithospheric structure reflect multistage tectonic evolution along the eastern Eurasian margin

Supporting Information:

Supporting Information may be found in the online version of this article.

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A Sharp Melt-Bearing Lithosphere–Asthenosphere Boundary Beneath the Korean Peninsula: Evidence From Teleseismic Receiver Functions

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Abstract The lithosphere–asthenosphere boundary (LAB) provides key constraints on how continental lithosphere responds to tectonic processes. Located between the East Asian cratonic blocks and the western Pacific subduction system, the Korean Peninsula (KP) provides an important setting to examine continental lithospheric evolution, yet seismic constraints on its LAB structure remain limited. We use P- and S-receiver functions to investigate crustal and lithospheric structure beneath the southern KP and adjacent volcanic islands. Beneath the southern KP, the LAB lies at ~70 km depth and is marked by a sharp transition zone (<20 km) and a pronounced shear-wave velocity (V_S) reduction (7%–10%), most plausibly explained by partial melt near the LAB. Thermodynamic modeling predicts melt fractions near the LAB, consistent with independent estimates from V_S reduction. These observations suggest that the southern KP represents the eastern continuation of a tectonically modified domain formed through long-term subduction-related thinning and removal of continental lithosphere. This modification varies laterally within the southern KP, where the southeastern block shows a deeper Moho, shallower LAB, and higher crustal P-to-S wave velocity ratio (V_P/V_S) than the northwestern block, indicating lithospheric overprinting by magmatic or thermal processes. Jeju Island retains inland-like Moho and LAB depths, but shows complex V_P/V_S variations and a broad lithosphere–asthenosphere transition. In contrast, Ulleung Island exhibits a shallow Moho, low V_P/V_S , and an exceptionally shallow LAB, reflecting additional lithospheric modification associated with Miocene backarc extension. Our results link the present-day lithospheric heterogeneity to the multi-stage tectonic evolution of the eastern Eurasian margin.

Plain Language Summary Imaging lithospheric thickness and structure helps reveal how tectonic processes have modified continental margins. We investigated the southern Korean Peninsula and the nearby volcanic islands of Jeju and Ulleung, where the lithosphere has been affected by Mesozoic thinning, Cenozoic backarc opening, and later volcanism. We used seismic waves from distant earthquakes together with thermodynamic modeling to image lithospheric structure. The lithosphere beneath the southern Korean Peninsula is relatively thin, with a lithosphere–asthenosphere boundary (LAB) at about 70 km depth, and its lower boundary is sharp. The decrease in shear-wave velocity across this boundary suggests that a small amount of molten rock may be present. Regional variations in lithospheric structure reflect the different tectonic histories of the peninsula and the adjacent volcanic islands. Jeju Island has crustal and lithospheric depths similar to those beneath the peninsula but a broader boundary, whereas Ulleung Island has a much shallower crust and lithosphere, reflecting additional modification related to Cenozoic backarc opening. These observations show that successive tectonic events and volcanism have produced distinct lithospheric structures along the eastern Eurasian margin.

1. Introduction

The Korean Peninsula (KP) is located along the eastern margin of the Eurasian Plate, transitioning between the East Asian cratonic blocks and the western Pacific subduction system (Figure 1a). This distinctive setting makes the KP an ideal locality to examine how continental lithosphere evolves from stable cratonic interiors toward an active plate margin. Extensive lithospheric thinning and craton destruction during the Early Cretaceous are well-documented beneath eastern China, where the lithosphere–asthenosphere boundary (LAB) has been imaged as shallow as ~60 km beneath the North China Craton (NCC; Chen et al., 2008, 2014; Y. Zhang et al., 2019) and South China Block (SCB; X. Li et al., 2024). Yet it remains a key question whether this pervasive lithospheric thinning extends to the eastern margin of the continent, and how the LAB varies laterally across the KP and

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toward its offshore margins. Furthermore, considering that Miocene backarc extension in the East Sea (Sea of Japan) has resulted in an exceptionally thin lithosphere (~45 km; Akuhara et al., 2021), understanding how the LAB varies laterally across the KP toward these offshore margins is essential to understanding the lithospheric evolution of the easternmost plate margin.

The present-day lithospheric architecture of the KP has been fundamentally shaped by a complex and episodic tectonic history. The southern KP comprises two Precambrian crystalline massifs, the Gyeonggi Massif (GM) and the Yeongnam Massif (YM), separated by the Okcheon Belt (OB) and Taebaeksan Basin (TB) which record medium-pressure metamorphism during 270–220 Ma (e.g., Cho et al., 2018). The Hongseong–Imjingang Belt (H–IB) flanks the northern GM; although its origin remains a subject of debate, the H–IB is often regarded as a potential eastern continuation of the collision system between the SCB and the NCC (e.g., S. W. Kim et al., 2026). The Gyeongsang Basin (GB) overlies the southeastern YM, which opened as a sediment basin during the Early Cretaceous driven by the paleo-Pacific Plate subduction (e.g., Chough & Sohn, 2010). In the Cenozoic, backarc extension associated with the opening of the East Sea (Sea of Japan) drove rifting and crustal thinning along the eastern margin of the KP, forming Miocene sedimentary basins such as the Yeonil Basin (YB) and intraplate volcanic islands including Ulleung Island and Dokdo in the East Sea (Figure 1b; M. Son et al., 2015; Van Horne et al., 2017). Cenozoic intraplate volcanism also produced Jeju Island to the south of the KP (Brenna et al., 2015). These tectonic processes collectively indicate that the KP preserves a spatially heterogeneous lithospheric structure shaped by multiple episodes of tectonic evolution.

The cumulative impact of this multistage evolution on the subcontinental lithospheric mantle is evidenced by petrological and geochemical constraints. Petrological evidence from mantle xenoliths indicates that the KP lithosphere is approximately 60–90 km thick (K.-H. Park, 2010), which is considerably thinner than the ~150 km thickness expected for its Precambrian lithospheric age. Geochemical studies from North Korean xenoliths further document significant Mesozoic lithospheric modification associated with decratonization (Yang et al., 2010). However, these xenolith constraints are spatially limited and cannot resolve lateral variations in LAB depth or continuity across the KP. Geochemical studies of Cretaceous igneous rocks on the KP (e.g., Im et al., 2021) and Cenozoic volcanic islands around the KP (e.g., D. Kim et al., 2021; W. Lee et al., 2025) independently indicate active lithosphere–asthenosphere interaction during magma generation. Together, these observations suggest metasomatic modification of the subcontinental lithospheric mantle along the eastern Eurasian plate margin.

Despite these petrological and geochemical indications, direct seismic constraints on the depth and velocity structure of the LAB beneath the KP and its surrounding volcanic islands remain limited. Receiver function studies have provided constraints on crustal thickness and P-to-S velocity ratio (V_P/V_S) beneath the KP, but these have been either limited to specific sub-regions (e.g., M. Kim, Lim, et al., 2026; Hwang et al., 2025) or based on sparse station networks (Chang & Baag, 2007), and none have attempted a comprehensive peninsula-wide analysis linking crustal properties to lithospheric thickness and LAB characteristics. Seismic tomography of the southern KP imaged a high-velocity anomaly beneath the west contrasting with low-velocities beneath the east, suggesting spatially heterogeneous lithospheric modification (H. Lee et al., 2026; Song et al., 2020). However, the intrinsic smoothing nature of tomography precludes reliable imaging of sharp discontinuities such as the LAB or any intra-lithospheric layering (e.g., mid-lithospheric discontinuity, MLD; Chen et al., 2014; Meng et al., 2021; Y. Zhang et al., 2019). As a result, the lithospheric architecture and evolution of the southern KP remain poorly constrained.

In this study, we present a comprehensive seismic investigation of the lithospheric structure beneath the southern KP, including its offshore volcanic islands of Jeju, Ulleung, and Dokdo, using an extensive data set from 300 broadband and accelerometer stations. By analyzing both teleseismic *P*- and *S*-receiver functions (PRFs and SRFs), we characterize the seismic discontinuity structure of the lithosphere from the crust to its lower boundary (LAB). Specifically, we (a) map crustal thickness and V_P/V_S across the southern KP, providing the crustal constraints necessary for interpreting SRF-derived LAB structure. We then (b) constrain LAB depth and sharpness using SRFs and forward modeling and (c) evaluate whether partial melt can account for the observed LAB sharpness by integrating thermodynamic melt fraction predictions with experimentally derived V_S –melt relationships. Collectively, these results provide high-resolution constraints on lithospheric structure and the origin of the sharp LAB along the eastern Eurasian margin.

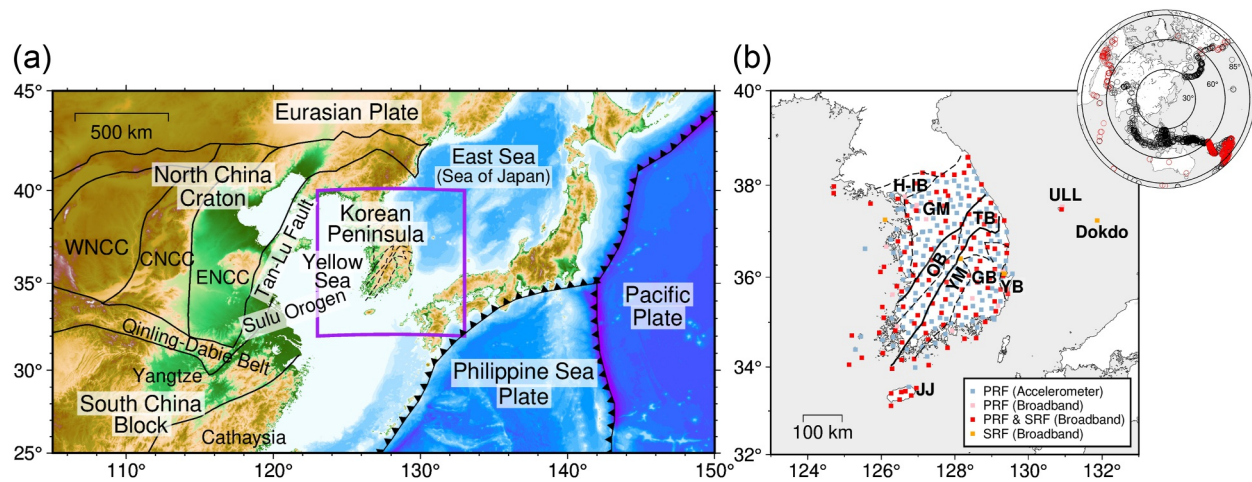


Figure 1. Tectonic framework of the study region and overview of the data set. (a) Tectonic setting of the Korean Peninsula (KP) and surrounding regions. The purple box outlines the study area shown in (b). Red triangles indicate locations of Cenozoic volcanism (Song et al., 2020). Solid lines indicate global tectonic province boundaries (Hasterok et al., 2022; H. Zhang et al., 2023), and dashed lines indicate geological boundaries within the southern KP (Kee et al., 2019). (b) Distribution of seismic stations and teleseismic events used in this study. Light blue squares indicate accelerometer stations, and pink squares indicate broadband stations for PRF analysis only. Red squares indicate broadband stations used for both PRF and SRF analyses, and orange squares indicate broadband stations for SRF analysis only. Black circles denote teleseismic events used for PRF calculations ($M_w \geq 5.8$; epicentral distances of 30° – 90°) that occurred between January 2010 and December 2022. Red circles denote teleseismic events used for SRF calculations ($M_w \geq 5.8$; epicentral distances of 60° – 85°) that occurred between January 2010 and October 2024. Abbreviations: ENCC, Eastern North China Craton; CNCC, Central North China Craton; WNCC, Western North China Craton; Yangtze, Yangtze Block; Cathaysia, Cathaysia Block; GM, Gyeonggi Massif; H-IB, Hongseong-Imjingang Belt; TB, Taebaeksan Basin; OB, Okcheon Belt; YM, Yeongnam Massif; GB, Gyeongsang Basin; YB, Yeonil Basin; ULL, Ulleung Island; JJ, Jeju Volcanic Island.

2. Data and Methods

2.1. Receiver Function Calculation

PRFs were calculated from teleseismic P-wave data recorded by 127 broadband seismometers and 173 accelerometers between January 2010 and December 2022, using data provided by the Korea Meteorological Administration (KMA) and the Korea Institute of Geoscience and Mineral Resources (KIGAM). We selected events with magnitudes ≥ 5.8 and epicentral distances of 30° – 90° (Figure 1b). Waveforms were cut from 30 s before to 90 s after the theoretical P arrival predicted by IASP91 (Kennett & Engdahl, 1991), corrected for sensor misorientation using station-specific angles (Y. O. Son et al., 2022), rotated to RTZ coordinates, and bandpass filtered between 0.05 and 1.0 Hz. PRFs were computed by iterative time-domain deconvolution of the vertical component from the radial component with a Gaussian filter width of 1.6 (Ligorria & Ammon, 1999; Xu & He, 2023) and normalized by the direct P amplitude. After quality control, 75,413 PRFs were retained for further analysis (Figure S1 and Text S1 in Supporting Information S1).

SRFs were calculated from teleseismic S-wave data recorded at 132 broadband seismometers between January 2010 and December 2024, using data provided by the KMA and KIGAM. Events with magnitudes ≥ 5.8 and epicentral distances of 60° – 85° were selected. Waveforms were bandpass filtered between 0.01 and 0.3 Hz, corrected for sensor misorientation, and rotated to LQT coordinates using theoretical S-wave incidence angles from IASP91 (Kennett & Engdahl, 1991; Y. O. Son et al., 2022). Waveforms were windowed relative to the theoretical S arrival using ± 100 s for the L component and -10 to $+35$ s for the Q component. SRFs were computed by iterative time-domain deconvolution of the L component by the Q component with a Gaussian filter width of 0.8 (Ligorria & Ammon, 1999; Xu & He, 2023), normalized by the S-wave amplitude, and inverted in both time and amplitude to follow the standard SRF polarity convention. Following the LQT-transform-based quality criteria from Shen et al. (2019), with additional visual inspection for island stations, we selected only SRFs showing a clear S-to-P converted phase from the Moho (Smp) at 3–5 s after the direct S arrival (Figure S1 in Supporting Information S1). Finally, 3,745 SRFs out of 61,482 initial records were retained owing to the inherently low signal-to-noise ratio of SRFs and the stringent requirement for a clear Smp phase.

2.2. H - κ Analysis Using P Receiver Functions

Crustal thickness (H) and crustal V_P/V_S ratio (κ) beneath each station were estimated using the H - κ stacking method applied to the PRFs (Zhu & Kanamori, 2000), based on the moveout of the P-to-S converted phase from the Moho (Pms) and its multiples in H - κ space. Crustal P-wave velocity (V_P) was assigned according to tectonic province following Chang and Baag (2005). A value of 6.2 km/s was used for stations in the GM and H-IB, 6.3 km/s for OB and TB, and 6.4 km/s for the YM and GB. For volcanic islands, V_P was set to 5.9 km/s for Ulleung Island (Chang & Baag, 2005) and 6.3 km/s for Jeju Island (Kang et al., 2021). The stacking was performed using the Ps and multiple phases, with weighting factors of 0.7 for Ps, 0.2 for PpPs, and 0.1 for PpPs + PsPs, respectively, and the reliability of the results was assessed by comparing predicted and observed phase arrivals.

At several stations, however, conventional H - κ stacking was insufficient because the RFs exhibited strong signatures of dipping interface, crustal anisotropy, and sediment reverberations. We therefore applied an extended H - κ - c stacking scheme of J. Li et al. (2019) to stations on Jeju Island, Ulleung Island, and the Jiri Mountain area (Figure S2 and Text S2 in Supporting Information S1), where such effects are particularly pronounced. PRFs were first moveout-corrected to a reference slowness of 0.06 s/km and stacked in 5° backazimuth bins to clarify azimuthal variations. We then performed a five-dimensional grid search to determine correction parameters that represent backazimuthal arrival time variations of the Pms phase and its multiples using $\cos\theta$ and $\cos 2\theta$ terms (J. Li et al., 2019). For Jeju Island, an additional sediment-related correction was applied to mitigate the influence of shallow sediment layer prior to the grid search (Kang et al., 2021; Yu et al., 2015). The optimal correction parameters were used to align phase arrivals before final H - κ stacking, with weights of 0.5, 0.4, and 0.1 for Ps, PpPs, and PpPs + PsPs, respectively (Figures S3 and S4 in Supporting Information S1). Uncertainties in H and κ were evaluated using the curvature of the stacking function (Zhu & Kanamori, 2000; Figure S5 in Supporting Information S1). Finally, the results were mapped using inverse-squared distance and uncertainty weighting with an effective radius of 8 km, corresponding to the horizontal distance of the Pms piercing point for a 30-km Moho.

2.3. Time to Depth Conversion of S Receiver Functions

The SRFs were converted from time to depth using a hybrid three-dimensional velocity model to account for lateral heterogeneity in the crust and mantle. The model combined the ambient-noise tomography model of S. Kim et al. (2016) above 40 km, which better resolves shallow crustal structure, with the full-waveform inversion model below 40 km for deeper lithospheric structure (Xi et al., 2024). Comparison with depth conversions using individual tomography models and IASP91 (Kennett & Engdahl, 1991) shows minor differences in absolute depth but preserves the first-order regional patterns (Figure S6 in Supporting Information S1). Depth uncertainties were estimated as the standard deviations of Moho and LAB depths from 1,000 bootstrap resamplings, each using 80% of the traces randomly resampled with replacement. Moho and LAB depth maps were then constructed using the same scheme as for the H - κ results, with effective radii of 24 and 66 km, respectively, based on S-to-P ray geometry for representative interface depths of 30-km Moho and 70-km LAB (Figures 2a and 3a).

2.4. Forward Modeling for S Receiver Functions

To constrain the velocity structure near the LAB, we performed forward modeling of stacked SRFs. Stations were grouped into four regions: the NW block, SE block, Jeju Island (JJ) and Ulleung Island (ULL). This grouping follows the regional crustal and lithospheric variations described in Sections 2.2 and 3.1–3.3. Within each group, we included as many usable SRFs as possible to obtain representative regional stacks, while prioritizing stations exhibiting clearly identifiable and relatively coherent Slp arrivals. Stations near the transitional crustal domains were excluded to avoid mixing structurally distinct regions within a single stack. For the NW and SE blocks, inland stations were preferentially selected to minimize the influence of contrasting coastal and offshore structures, considering the relatively large station-to-piercing-point distances of the SRFs. For JJ, stations distributed along the coast were selected, excluding those around Mt. Halla.

Before stacking, individual SRFs were moveout-corrected to a reference slowness of 0.1 s/km to enhance the coherence of Moho- and LAB-converted phases. Forward modeling was conducted using the one-dimensional waveform modeling code RAYSUM (Frederiksen & Bostock, 2000). For the NW block, SE block, and ULL, the average crustal V_P values were adopted from the PRF study of Chang and Baag (2005), whereas the crustal V_P for JJ was taken from Kang et al. (2021). For each region, crustal V_S was determined from the prescribed V_P/V_S

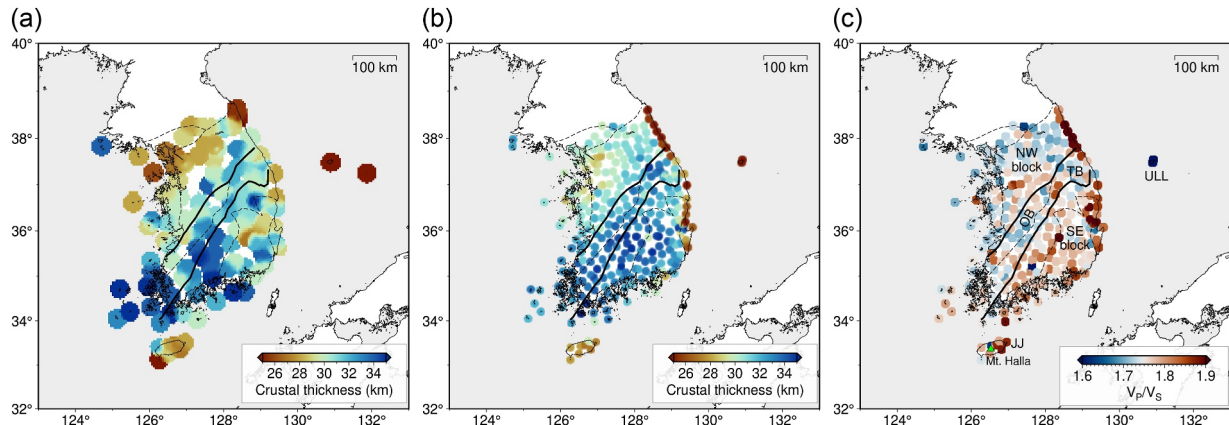


Figure 2. Crustal thickness and P - to S -wave velocity ratio (V_P/V_S) constrained from PRF and SRF analyses. (a) Crustal thickness from depth-converted SRFs. (b) Crustal thickness and (c) V_P/V_S from the H - κ stacking method (J. Li et al., 2019; Yu et al., 2015; Zhu & Kanamori, 2000). The NW comprises the GM and H-IB, and the SE block comprises the YM and GB. Abbreviations are defined in Figure 1.

ratio, which was chosen to reproduce the observed amplitude of the Smp phase in the stacked SRFs within H - κ derived crustal thickness and V_P/V_S range. Mantle V_P and V_S were initially determined based on the global model of IASP91 and were then adjusted to match the observed Smp amplitude. In the preferred model, mantle V_P and V_S beneath the SE block and ULL were reduced by 1.5% and 1.0%, respectively, relative to those beneath the NW block. Crustal and mantle densities were calculated from V_P using the empirical relation of $\rho = 770 + 0.32V_P$ (Berteussen, 1977), where ρ is density in kg/m^3 and V_P in m/s. The resulting model parameters are summarized in Table 1.

The modeling focused on constraining the magnitude of the velocity reduction at the LAB and the thickness of the lithosphere-asthenosphere transition zone. Within the transition zone, both V_P and V_S were assumed to decrease linearly (Figure 4a). For each group, synthetic SRFs were generated for models with varying velocity drops (3%–17%) and transition thicknesses (0–40 km), assuming a reference slowness of 0.1 s/km. For each transition thickness, the best-fitting velocity reduction was selected by minimizing the amplitude misfit between the observed and synthetic S-to-P converted phases from the LAB (Slp).

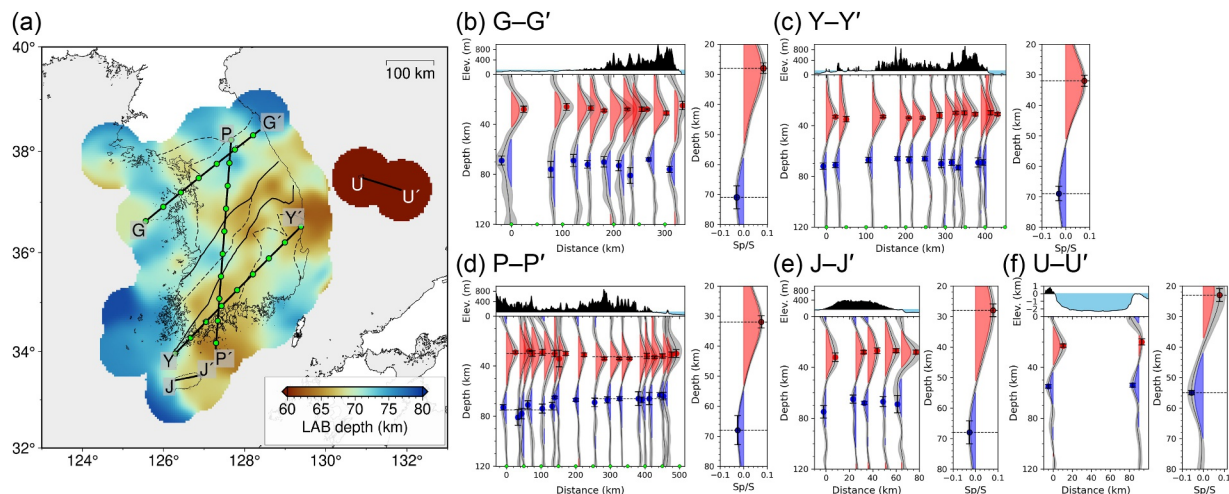


Figure 3. Lithosphere-asthenosphere boundary (LAB) depth constrained by SRF analysis. (a) LAB depth map and locations of the SRF profiles. (b–f) SRF profiles obtained by station-based stacking along (b) G–G', (c) Y–Y', (d) P–P', (e) J–J', and (f) U–U'. In panels (b–f), red and blue dots mark the depths of the Moho and LAB, respectively, derived from depth-converted S -to- P phases. Error bars represent one standard deviation estimated from 1,000 bootstrap resamplings. Gray shaded envelopes indicate the uncertainty of the stacked SRFs derived from the same bootstrap procedure. The profile-averaged SRF, obtained by stacking all SRFs within each profile, is shown on the right side of each panel. Green dots in panels (a–d) indicate locations sampled at 50-km intervals along each profile.

Table 1

Best-Fit Parameters for SRF Modeling for Each Station Group, Including Moho Depth, Lithosphere–Asthenosphere Boundary (LAB) Depth, Crustal and Lithospheric V_P and V_P/V_S , LAB Velocity Reduction, and LAB Transition Thickness

	Moho depth (km)	LAB depth (km)	V_P (crust) (km/s)	V_P/V_S (crust)	V_P (lithosphere) (km/s)	V_P/V_S (lithosphere)	V_S drop at LAB	LAB transition thickness
NW block	30	71	6.2	1.71	8.0	1.8	6.8%–9.8%	<20 km (sharp)
SE block	32	66	6.4	1.77	7.88	1.8	7.0%–9.6%	<20 km (sharp)
ULL	23	57	5.9	1.64	7.92	1.73	11.5%–12.5%	<20 km (sharp)
JJ	26	66	6.3	1.82	8.0	1.83	7.5%–11%	>20 km (broad)

2.5. Thermodynamic Modeling of Melt Fraction

The melt fraction at the LAB was estimated using the thermodynamic model pHMELTS (Asimow et al., 2004; Ghiorso et al., 2002), as implemented in the alphaMELTS v.1.9 package (Smith & Asimow, 2005). Pressure–temperature conditions were assigned according to the LAB depth estimated from SRF analysis for each group (Table 1). At depths below LAB, temperatures were constrained to follow the adiabatic geotherm of the Depleted MORB Mantle (DMM; Workman & Hart, 2005) with a potential temperature (T_p) of 1300°C, consistent with the definition of the thermal LAB (Herzberg et al., 2007; O'Reilly & Griffin, 2010). Above the LAB, temperatures were assumed to decrease linearly from the LAB to 0°C at the surface. To evaluate the effect of thermal uncertainty, additional models were calculated using T_p values of 1280 and 1320°C. The results are shown as dashed lines in Figure 5.

The major element and H₂O compositions of the asthenosphere were based on the DMM values from Workman and Hart (2005), with the redox state fixed at $\Delta\text{FMQ-1}$ (Frost & McCammon, 2008). To account for regional variations in the lithospheric mantle composition, sample BR18 (spinel Cr# = 9.2, molar Cr/(Cr + Al) $\times$ 100), one of the most fertile mantle xenoliths in the southern KP (S. H. Choi & Kwon, 2005) was used as a representative lithospheric mantle composition. The lithospheric mantle redox state was also set to $\Delta\text{FMQ-1}$, consistent with estimates from Korean mantle xenoliths ($\Delta\text{FMQ-1.5}$ – $\Delta\text{FMQ-0.5}$; G. Park et al., 2018). The lithospheric H₂O content was fixed at 500 ppm, corresponding to the water contents inferred for Cenozoic Korean magmas (1.5–2.0 wt.% H₂O; J. Park et al., 2022; see Text S3 in Supporting Information S1).

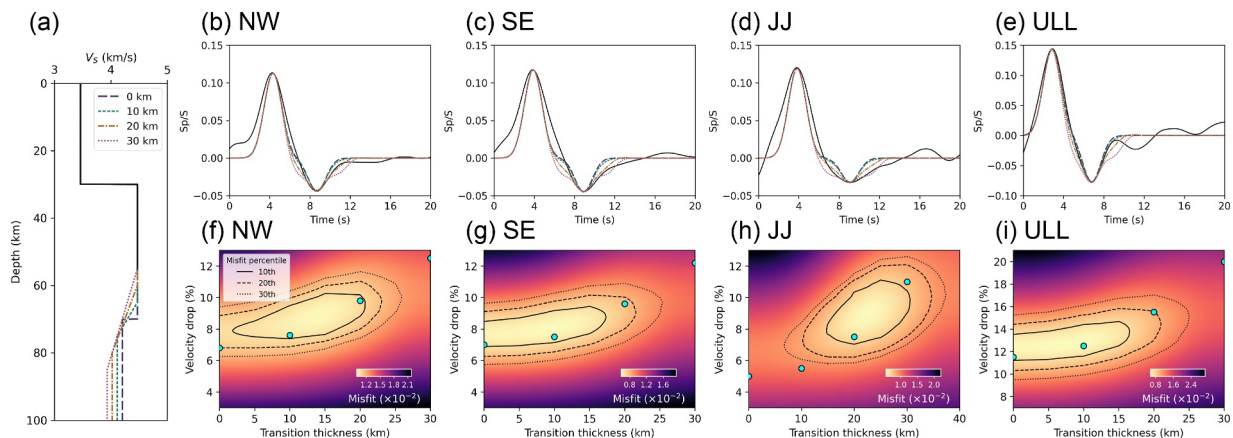


Figure 4. SRF modeling results for quantifying the sharpness of the lithosphere–asthenosphere boundary (LAB). (a) Representative velocity models used for SRF forward modeling, showing a sharp Moho and LABs with transition thicknesses of 0, 10, 20, and 30 km paired with velocity drops of 6%, 8%, 10%, and 12%, respectively. Panels (b)–(e) show the observed stacked SRFs (black solid lines) and synthetic SRFs (colored lines) for (b) the NW block, (c) SE block, (d) JJ, and (e) ULL, with line colors corresponding to the transition thicknesses shown in panel (a). Misfits between observed and synthetic SRFs for each region are shown in panels (f)–(i), constructed by varying the LAB transition thickness in 5-km increments and the velocity drop in 0.1% increments. Black solid, dashed, and dotted contours delineate the lowest 10%, 20%, and 30% of the misfit distribution, respectively. Cyan circles indicate the preferred velocity reductions for LAB transition thickness of 0, 10, 20, and 30 km, corresponding to the models shown in panels (b)–(e).

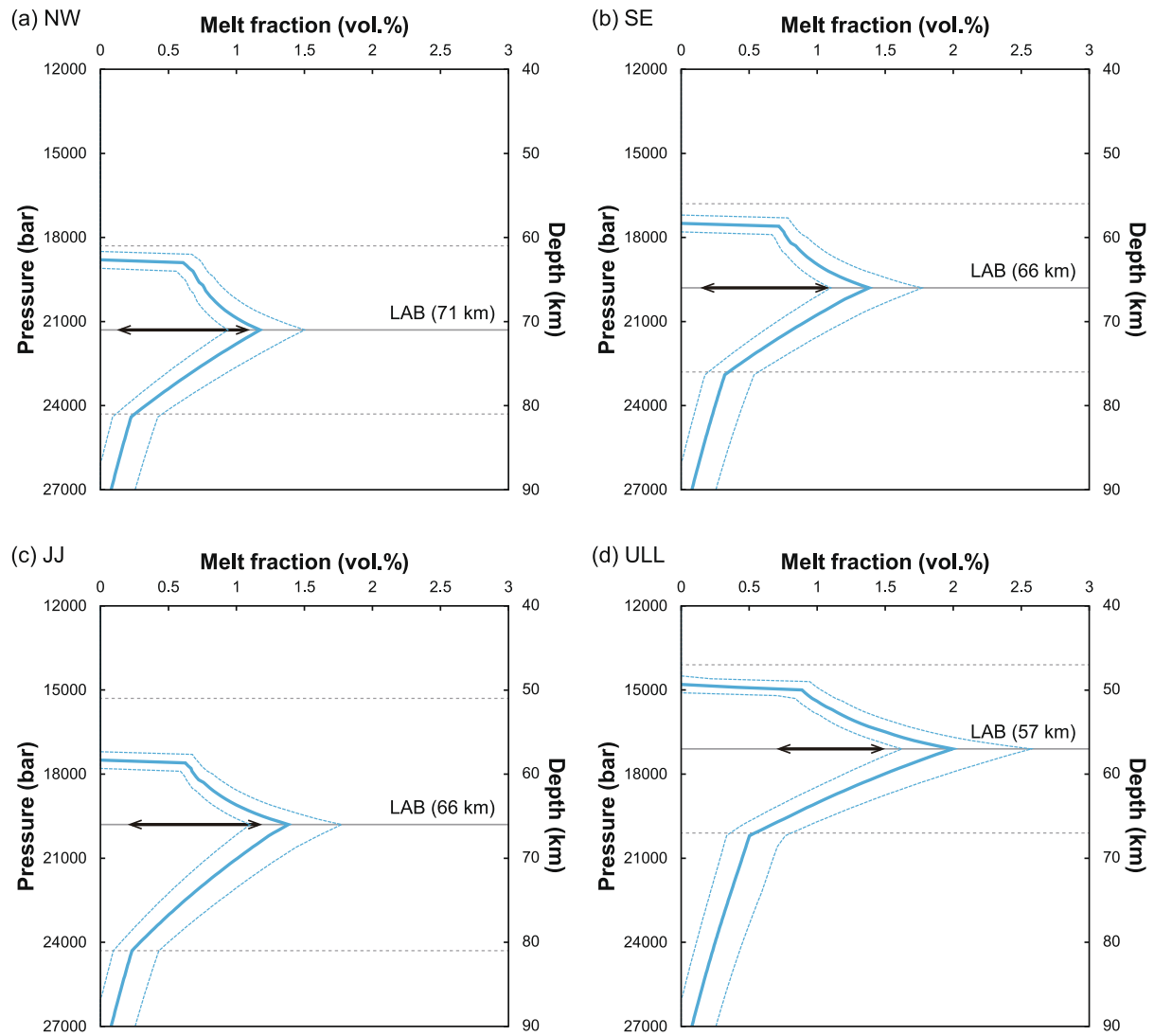


Figure 5. Melt fraction estimates for (a) the NW block, (b) SE block, (c) JJ, and (d) ULL. The blue solid line represents melt fraction estimates based on thermodynamic modeling assuming a potential temperature of 1300°C, while blue dashed lines represent results for 1280°C and 1320°C. The lithosphere–asthenosphere boundary (LAB) depth (gray solid lines) and transition thickness (gray dashed lines) assumed for each group follow the best-fit parameters in Table 1; a transition thickness of 20 km was assumed for the NW, SE, and ULL groups, and 30 km for JJ. Black arrows indicate the range of estimated melt fraction from Hammond and Humphreys (2000), based on the best-fit V_S drops at the LAB listed in Table 1.

The model simulates a series of batch melting events with 100 mbar pressure increments across the pressure range corresponding to the transition zone of each group (Table 1; Figure 4). Because the relative contributions of asthenospheric and lithospheric mantle within the transition zone are poorly constrained, a linear mixing relationship between the two end-members was assumed over the corresponding pressure interval (see Text S3 in Supporting Information S1 for alternative schemes). The bulk melt fraction at each step was then calculated as the weighted average of the melt fractions produced from the asthenospheric and lithospheric components.

For comparison with the thermodynamic modeling results, we independently estimated melt fractions based on the V_S -reduction–melt-fraction trends of Hammond and Humphreys (2000), derived from finite element calculations incorporating experimentally observed melt geometries. Their study considered two melt-containment models: one assuming film-like melt geometries at all melt fractions and another allowing a transition from triple-junction tubules to film-dominated geometries with increasing melt fraction. The V_S reductions obtained from the forward modeling of the SRFs were then used to estimate the range of melt fractions based on these two melt-containment models.

3. Results

3.1. Crustal Thickness

Both PRFs and SRFs show clear Moho conversions across the KP (Figure S1 in Supporting Information S1). The PRF- and SRF- based crustal thickness are mutually consistent across nearly all stations (Figures 2a and 2b), with only one station exceeding the 95% confidence interval of the Z-test (Text S4 and Figure S9 in Supporting Information S1), demonstrating consistency between the independent PRF- and SRF-based Moho estimates.

Crustal thickness derived from H - κ stacking of PRFs exhibit systematic regional variations across the study area (Figure 2b). In the GM and H-IB region, the average crustal thickness is 30.8 ± 1.0 km, whereas in the YM and GB region it is thicker, averaging 32.5 ± 1.8 km, with values reaching up to ~ 35 km in the western YM. We note that Monte Carlo uncertainty propagation confirms that the mean crustal thickness beneath GM and H-IB is significantly thinner by 1.7 km than that beneath YM and GB, with 95% confidence interval of 0.9–2.5 km, even when station-specific uncertainties are taken into account (Text S5 and Figure S10a in Supporting Information S1). The OB and TB regions show comparable crustal thickness to YM and GB, with an average of 33 ± 1.0 km. In contrast, the crustal thickness becomes significantly thinner toward the eastern coastal region of the KP, where an average thickness of 25.4 ± 2.5 km is observed (Figure 2b). An even thinner crustal thickness is found beneath ULL, with a thickness of 20.3 ± 0.4 km. Beneath Jeju Island (JJ), the crustal thickness is estimated to be 27.9 ± 0.8 km.

3.2. Crustal V_P/V_S Ratio

The spatial distribution of crustal V_P/V_S derived from H - κ stacking of PRFs shows clear regional variations across the KP (Figure 2c). Relatively low V_P/V_S ratios of 1.74 ± 0.03 are observed in the GM and H-IB, whereas higher values of 1.78 ± 0.05 are found in the YM and GB. This contrast is particularly evident across the OB and TB regions. Monte Carlo uncertainty propagation indicates that V_P/V_S is higher in the YM and GB than in the GM and H-IB by 0.04 on average, with 95% confidence interval of 0.02–0.05 (Text S5 and Figure S10b in Supporting Information S1). Along the eastern coastal region of the KP, elevated V_P/V_S ratios of 1.86 ± 0.08 are observed (Figure 2c). On JJ, high V_P/V_S ratios of 1.81 ± 0.05 are observed along the coastal areas, whereas significantly lower values of 1.63 ± 0.03 are found near Mt. Halla, corresponding to the central part of the volcanic island. Similarly, low V_P/V_S ratios of 1.6 ± 0.04 are observed on ULL.

3.3. LAB Depth

Stacked SRFs from epicentral distance binning show a coherent negative phase at ~ 8 – 10 s, indicating a broadly consistent LAB arrival across epicentral-distance bins (Figure S11 in Supporting Information S1). Depth migration of SRFs yields an average LAB depth of 69.9 ± 4.5 km beneath the KP (Figure 3a). The LAB is laterally continuous across the peninsula, with depth variations limited to ~ 3.4 km.

A regional difference is also apparent between the G-G' and Y-Y' profiles, which yield mean LAB depths of 72.7 ± 4.0 km and 69 ± 2.3 km, respectively (Figures 3b and 3c). This difference is robust within station-specific uncertainties, with Monte Carlo uncertainty propagation indicating a 3.4 km shallower LAB beneath the YM and GB relative to the GM and H-IB on average (95% confidence interval: 2.0–4.9 km; Text S5 and Figure S10c in Supporting Information S1). The most pronounced lateral variation is observed along P-P' profile across the OB and TB that separate two contrasting domains: the GM and H-IB region, characterized by a relatively thin crust (30.0 ± 1.9 km) and deep LAB (74.8 ± 3.5 km), and the YM and GB region, characterized by a thicker crust (32.5 ± 1.4 km) and shallower LAB (66.0 ± 1.7 km) (Figure 3d, dashed lines).

In the J-J' profile across JJ, the mean LAB depth is 69.0 ± 6.3 km, with a relatively larger standard deviation than those observed in other regions, likely reflecting the complex volcanic structure beneath the island (Figure 3e). In contrast, the U-U' profile shows that the LAB beneath ULL and Dokdo is significantly shallower (54.5 ± 0.7 km) than the KP average (69.9 ± 4.5 km) (Figure 3f).

Based on the coherent contrasts in crustal thickness, crustal V_P/V_S , and LAB depth, the GM + H-IB and YM + GB regions can be grouped into two distinct crust-lithosphere domains. Hereafter, we refer to these domains as the NW block and SE block, respectively, with OB and TB marking their boundary.

3.4. LAB Sharpness

Forward modeling of SRFs was performed for four representative groups (the NW block, SE block, JJ, and ULL) to constrain the velocity reduction and transition thickness at the LAB (see Section 2.4). The results show a trade-off between transition thickness and the required velocity reduction: a thinner LAB transition requires a smaller velocity reduction to reproduce the observed S/p amplitude, whereas a broader transition requires a larger reduction (Figure 4).

The LAB beneath the NW block, SE block, and ULL is characterized by a sharp transition of ≤ 20 km, whereas the LAB beneath JJ shows a more gradual transition. For a sharp LAB transition (0 km thickness), the required S -wave velocity reductions are 6.8% for the NW block, 7.0% for the SE block, 5.0% for JJ, and 11.5% for ULL. These values increase systematically with transition thickness, reaching 12.5%, 12.2%, 11%, and 20% for a 30 km transition, respectively (Figures 4b–4e). Given the vertical resolution limit imposed by the dominant frequency of the SRFs (~ 0.1 Hz), models with transition thicknesses ≤ 20 km cannot be distinguished from one another. For the NW block, SE block, and ULL, the 30 km transition model yields a systematically poorer fit than models with transition thicknesses of ≤ 20 km (Figures 4f, 4g, and 4i), constraining the velocity reductions to 6.8%–9.8% (NW block), 7.0%–9.6% (SE block), and 11.5%–12.5% (ULL), respectively (Table 1). In contrast, the JJ group shows a different pattern: the best-fit model corresponds to a broader transition thickness than 20 km (Figure 4h) with a 7.5%–11% velocity reduction (Table 1), consistent with the broader S/p phase observed in the JJ stacked SRFs compared to the other groups, suggesting a more gradual LAB transition.

3.5. Melt Fraction at the LAB

Melt fraction estimates derived from two independent approaches, thermodynamic modeling (Smith & Asimow, 2005) and the rock physics-based V_S –melt relationship (Hammond & Humphreys, 2000), are broadly consistent across all groups. Thermodynamic modeling yields melt fraction profiles across the depth range of 40–90 km (Figure 5). Owing to its hydrous nature, the lithospheric mantle exhibits substantially higher melt fractions than the asthenospheric mantle. In the combined bulk model, the melt fraction remains below 0.75 vol.% at depths beneath the lithosphere–asthenosphere transition zone. Within this zone, melt fractions progressively increase with decreasing pressure, reaching maximum values of 0.9–2.6 vol.% near the LAB as the contribution from the hydrous lithospheric mantle becomes increasingly significant. The melt fraction above the LAB gradually decreases with decreasing pressure, reflecting conductive cooling within the mantle. The maximum melt fractions for the SE block and JJ (1.1–1.8 vol.%) are slightly higher than those for the NW block (0.9–1.5 vol.%). Notably, the ULL exhibits a substantially higher melt fraction of 1.6–2.6 vol.% than the other groups. An independent estimate derived from the observed V_S drops at the LAB (Table 1) using Hammond and Humphreys (2000) also yields the highest value of 0.70–1.29 vol.% for ULL, while melt fractions for the other groups are broadly similar (Figure 5; 0.12–1.10 vol.% for the NW block; 0.14–1.09 vol.% for the SE block; 0.20–1.19 vol.% for JJ).

4. Discussion

4.1. A Pervasively Thinned Lithosphere With a Sharp LAB Beneath the Southern Korean Peninsula

The lithosphere beneath the southern KP is characterized by significant thinning and shares key characteristics with the tectonically modified lithosphere of eastern China, where changes in the geometry of paleo-Pacific subduction are thought to have driven widespread lithospheric thinning and craton destruction during the Cretaceous (S. H. Lee et al., 2023; Wu et al., 2019; H. Zhang et al., 2023). The SRF images support this interpretation by revealing a relatively uniform LAB at ~ 70 km depth, comparable to those reported in eastern China (e.g., X. Li et al., 2024; Y. Zhang et al., 2019), with no clear evidence for an MLD (Figures 3, 4, and 6b). This depth is markedly shallower than LAB depths reported for relatively preserved cratonic lithosphere (~ 150 – 200 km or greater), including LAB depths of >200 km beneath the western NCC (Chen, 2010; Chen et al., 2014). Such characteristics are consistent with the evolution of a tectonically reworked continental lithosphere, where extensive lithospheric thinning removes or obscures pre-existing intra-lithospheric structures such as an MLD (X. Li et al., 2024; Meng et al., 2021; Sun et al., 2020).

The LAB beneath the southern KP is not only shallow but also seismically sharp. Similar sharp LAB structures have been reported beneath eastern China, where the lithosphere is markedly thinned and bounded by a relatively

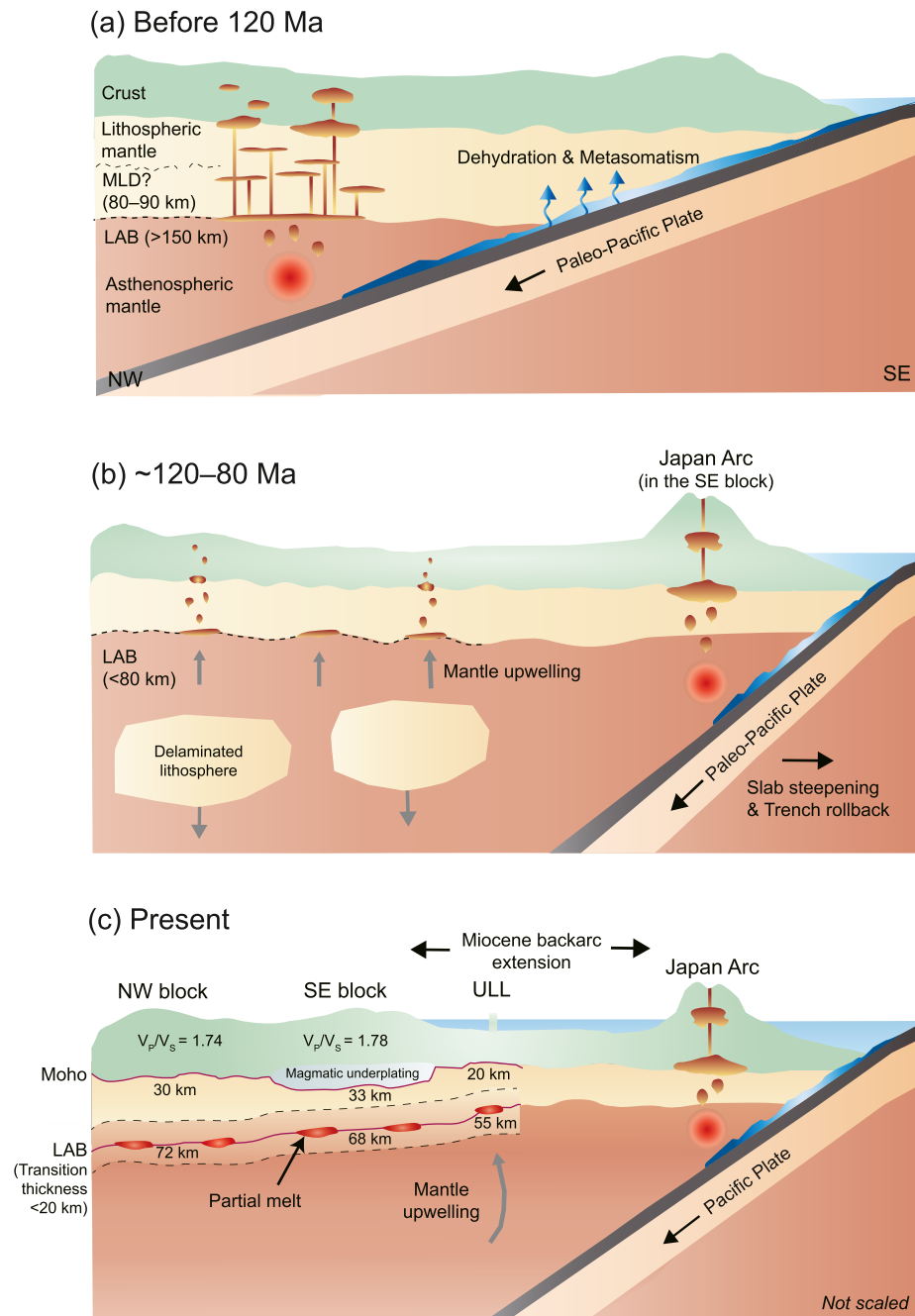


Figure 6. Conceptual model for the tectonic evolution of the lithosphere beneath the Korean Peninsula. (a) Prior to 120 Ma, paleo-Pacific subducted beneath the Eurasian margin at a shallow angle. Magmatism occurred far from the trench, mainly in the region of present-day eastern China (e.g., Wu et al., 2019). During this process, slab-derived fluids migrated into the overlying lithospheric mantle, likely causing metasomatism and increasing water content. (b) Subsequent slab steepening and trench rollback may have promoted lithospheric delamination along a mechanically weak mid-lithospheric discontinuity/weak layer (MLD), leading to modification of the pre-existing lithospheric mantle. The associated asthenospheric upwelling may have induced melting, promoting localized magmatism (e.g., Im et al., 2021). (c) Continued Pacific slab rollback drove Miocene backarc extension, resulting in separation of the Japan Arc from the KP and termination of arc magmatism beneath the peninsula. The present-day lithospheric structure of the KP therefore reflects the cumulative effects of arc-related modifications, lithospheric delamination, and Miocene backarc extension.

sharp LAB. Beneath the Tanlu Fault Zone, the LAB is characterized by a 3%–7% S -wave velocity drop over ≤ 10 km (Chen et al., 2006). Across the northeastern NCC, the LAB transition is similarly narrow, generally on the order of 10–20 km with a velocity decrease of $\sim 6.5\%$ (Chen et al., 2008; Meng et al., 2021). Beneath the eastern NCC, the LAB transition is as sharp as ~ 2 km, as inferred from P -wave autocorrelogram analysis (Sun et al., 2020). The LAB beneath the southern KP is likewise characterized by a sharp boundary with a transition thickness of < 20 km, with V_S reductions ranging from 6.8% to 9.8% depending on the assumed transition thickness (see Section 3.4). These values are broadly comparable to those reported in eastern China, suggesting that a sharp, high-contrast LAB may be a common feature of tectonically modified lithosphere along the eastern margin of the Eurasian Plate.

4.2. Partial Melt as the Origin of the Sharp LAB

The sharp LAB observed across the study area is best explained by the presence of partial melt at the base of the lithosphere. The sharpness of the lithosphere–asthenosphere transition can be influenced by multiple factors including temperature, chemical composition, grain size, and melt content (Faul & Jackson, 2005; Fischer et al., 2010), but each of these alternatives alone appears insufficient to explain the observed velocity contrast. A purely thermal origin cannot fully account for the observed V_S drop, as mantle flow models predict thermal gradients of less than $5^\circ\text{C}/\text{km}$ at the base of the lithosphere, with an upper bound of $\sim 10^\circ\text{C}/\text{km}$ (Rychert et al., 2005). Such gradients can explain a maximum V_S drop of only 3.8%–7.5% across a 20 km transition, insufficient to account for the best-fit velocity reductions of 9.8% (NW block), 9.6% (SE block), and 12.5% (ULL) at the same transition thickness. Chemical depletion of the lithosphere has a negligible effect on V_S ($< 1\%$ – 2%) across the LAB (Schutt & Leshner, 2006), while hydration could account for at most $\sim 1.3\%$ – 4.3% V_S decrease (Rychert et al., 2005, 2007), collectively insufficient to explain the observed V_S drop of 9.8%–12.5%. Vertical differences in grain size across the LAB may also contribute to the velocity contrast; however, producing a sufficiently sharp grain size contrast at the LAB would require a sudden increase in shear stress with depth, which is inconsistent with mantle flow models where shear stress generally decreases gradually with depth (Rychert et al., 2005). Partial melt provides the most plausible explanation for the observed LAB sharpness, as melt fractions of 1 vol.% can reduce V_S by at least 7.9% in realistically shaped inclusions (Hammond & Humphreys, 2000). This interpretation is further supported by the thermodynamic modeling results (Section 3.5), which independently predict melt fractions of ~ 0.9 – 1.8 vol.% for the southern KP and up to ~ 2.6 vol.% beneath ULL. Such estimated melt fractions are consistent with a hydrous lithospheric mantle, potentially reflecting prior fluid-related metasomatism during shallow-angle paleo-Pacific subduction that increased water contents and lowered the mantle solidus (Figure 6a).

The melt-bearing, sharp LAB observed beneath KP is not unique to this region, but rather appears to be a recurrent feature across a range of continental tectonic settings. Beneath eastern China, sharp LAB structures have similarly been interpreted as reflecting a melt-rich layer at the lithospheric base (Chen et al., 2006; Meng et al., 2021). In the western United States, SRF imaging reveals a sharp LAB characterized by an average velocity reduction of $\sim 10\%$ across a 30-km-thick transition, which has been interpreted to require ponded partial melt beneath the lithosphere (Hopper & Fischer, 2018). Sharp velocity contrasts at the LAB that cannot be explained by purely thermal gradients have further been documented beneath eastern North America, where partial melt or volatile enrichment has been invoked as a contributing factor (Rychert et al., 2005), and beneath Phanerozoic eastern Australia (Ford et al., 2010). Together, these global observations suggest that melt-bearing, sharp LABs might represent a widespread feature of continental lithosphere across diverse tectonic settings.

4.3. Regional Variations in Lithospheric Characteristics Influenced by Arc Magmatism Since Early Cretaceous

Combined analyses of crustal thickness, V_P/V_S ratio and LAB depth reveal systematic differences between the NW and SE blocks. The SE block is characterized by a thicker crust, shallower LAB, and higher V_P/V_S than the NW block (Figures 2 and 3; Figure S10 in Supporting Information S1). The pattern is consistently observed from independent PRF- and SRF-based analyses, supporting its reliability.

The contrasting crustal thickness and LAB depth patterns suggest distinct lithospheric evolution beneath the NW and SE blocks. The thicker crust and elevated V_P/V_S in the SE block are consistent with mafic additions to the lower crust (Hacker et al., 2015), likely associated with voluminous Cretaceous arc magmatism concentrated in

southeastern parts of the KP (Figure 6b; Chough & Sohn, 2010; Im et al., 2021). Magmatic underplating may have contributed to crustal thickening (Thybo & Artemieva, 2013). The stacked *Smp* amplitude is slightly larger in the NW block than in the SE block, with a difference of $\sim 2\%$ (Figures 4b and 4c). To reproduce this small amplitude difference together with the *Smp* arrival time, the mantle velocities beneath the SE block were set to be 1.5% lower than those beneath the NW block in the forward modeling (Table 1). This parameterization is consistent with previous tomographic images showing low-velocity anomalies beneath eastern KP at starting depths of ~ 50 – 60 km, possibly related to asthenospheric upwelling. Such upwelling could also account for the slightly shallower LAB beneath the SE block (H. Lee et al., 2026; Song et al., 2020). In contrast, the NW block exhibits a thinner crust, thicker lithosphere, and lower V_p/V_s ratios, consistent with relatively more felsic crustal compositions (Christensen, 1996; Hacker et al., 2015) and a lesser degree of magmatic modification.

In contrast, the sharpness of the LAB shows only limited regional differences. The resolvable thickness of the LAB transition from SRFs is constrained by the dominant frequency content, corresponding to an effective vertical resolution of ~ 20 km. Synthetic modeling results indicate that transition thicknesses smaller than this scale cannot be reliably distinguished, and little variation in waveform broadening is observed for transition thicknesses of 0–20 km (Figure 4). Accordingly, relatively sharp LABs (transition thickness < 20 km) are generally inferred beneath both the NW and SE blocks (Figures 4b, 4c, 4f, and 4g; Table 1). Within this constraint, the estimated velocity reductions across the LAB are 6.8%–9.8% in the NW block and 7.0%–9.6% in the SE block (Table 1). This similarity in velocity reductions is consistent with the comparable melt fractions estimated for the two regions. Thermodynamic modeling results exhibit broadly similar melt fraction estimates in the SE region and NW region (1.1–1.8 vol.% vs. 0.9–1.5 vol.%; Figures 5a and 5b).

4.4. Complex Lithospheric Properties Beneath Jeju Volcanic Island

The crustal thickness and LAB depth beneath JJ are broadly comparable to those beneath the inland regions, consistent with its location within the continental lithosphere (Brenna et al., 2015; Sohn et al., 2008). However, the crust and uppermost mantle beneath JJ appear more heterogeneous, reflecting additional influence of Quaternary volcanic activity. The crustal thickness beneath JJ is estimated to be ~ 28 km, consistent with previous estimates (Kang et al., 2021). The crustal V_p/V_s ratios additionally show spatial variations across JJ, with higher values along the coastal regions reflecting basaltic compositions, and lower values near Mt. Halla corresponding to the presence of more evolved, trachytic-dominated volcanic rocks (Kang et al., 2021). These observations collectively suggest that crust beneath JJ has undergone complex magmatic evolution.

The LAB beneath JJ, however, appears more gradational and heterogeneous than those beneath the inland region, as suggested by its broader transition zone (> 20 km) and larger depth standard deviation (Figures 3e and 4h). This broad LAB transition may reflect a more complex LAB system, possibly related to its multiple volcanic events. A prominent upper mantle low-velocity anomaly at ~ 50 – 60 km depth has been reported beneath JJ (Song et al., 2018). Considering ~ 20 km vertical resolution of SRFs, such a low-velocity structure near the LAB may broaden the LAB-related signal and make the lithosphere–asthenosphere transition appear more complex and gradational. The estimated velocity reduction across the LAB is also comparable to or locally larger (7.5%–11%) than that of the inland regions (6.8%–9.8%), implying that slightly higher melt may be more broadly distributed over a wider depth interval rather than concentrated within a narrow boundary. Although this elevated melt fraction for JJ is not reproduced in thermodynamic modeling (Figure 5c), this discrepancy may reflect the simplified lithospheric compositions assumed in this study. The lithospheric mantle beneath JJ may have been more extensively modified by melt-related metasomatism during multiple Cretaceous and Cenozoic magmatic episodes than assumed in the thermodynamic models used in this study.

4.5. Crustal and Lithospheric Modification Beneath the Eastern Korean Peninsula and Ulleung Island During Miocene Backarc Extension

Along the eastern margin of the KP, the crustal thickness is systematically shallower (at depths of ~ 25 km), and crustal V_p/V_s is higher (~ 1.86) compared to the inland regions, in both the NW and SE blocks (Figures 2b and 2c). This spatially coherent pattern suggests a later crustal modification that overprints the pre-existing structural contrast between the NW and SE blocks related to Cretaceous arc magmatism. This crustal thinning is even more prominent beneath ULL, where both the crust and lithosphere are thinner than beneath the eastern margin. These regional variations suggest crustal and lithospheric thinning related to extensional deformation during the

Miocene backarc opening, which likely played an important role in shaping the present-day crustal and lithospheric structure of the region (e.g., Hwang et al., 2025; M. Kim, Lim, et al., 2026; S. Park & Hong, 2024). This backarc opening occurred during the Miocene (23–17 Ma) as a result of Pacific plate rollback, leading to the separation of the Japanese arc from the KP (Figure 6c; Akuhara et al., 2021; Van Horne et al., 2017). The lithospheric structures beneath ULL are markedly different from those beneath JJ, suggesting that these two volcanic islands were modified by Cenozoic intraplate volcanism under different tectonic settings.

A key distinction between the eastern margin of KP and ULL is the opposite character of their crustal V_P/V_S , which indicates different styles of Cenozoic magmatic modification. The relatively high V_P/V_S along the eastern margin may reflect mafic addition to the crust during the Miocene backarc extension, as supported by the occurrence of basaltic volcanic rocks in parts of the coastal regions (e.g., H.-O. Choi et al., 2013). In contrast, the low V_P/V_S beneath ULL is difficult to reconcile with simple mafic underplating or basaltic modification. Instead, it may reflect compositional heterogeneity associated with repeated basaltic to trachytic–phonolitic volcanism (Brenna et al., 2014). In addition, gas- or fluid-bearing fractured rocks (W. Lee et al., 2025) may contribute to the observed low V_P/V_S , because changes in V_P/V_S depend on pore geometry and elastic properties of the fluid-rock system (Brantut & David, 2019).

Post-opening volcanism beneath ULL is likely linked to the high melt fraction at the LAB. The thermodynamically estimated melt fraction locally exceeds ~ 2 vol.% (Figure 5d), which is similar to or greater than the threshold for stable melt accommodation along grain boundaries (2–3 vol.%; Faul, 1997; ten Grotenhuis et al., 2005). Such conditions indicate that the melt may become interconnected and potentially mobilized in response to tectonic perturbations. This may indicate a higher potential for melt extraction than beneath the inland regions and JJ. The relatively high heat flow observed in the ULL (>120 mW/m²; H. Kim et al., 2019), compared with the KP average (60 ± 11 mW/m²; H. C. Kim & Lee, 2007), supports this interpretation.

5. Conclusions

Combined analyses of P- and S-receiver functions and thermodynamic modeling provide new constraints on the lithospheric architecture beneath the southern KP and adjacent volcanic islands and on its role in the tectonic evolution of the eastern Eurasian margin. Our results reveal a relatively thin lithosphere, with the LAB located at ~ 70 km depth and marked by a sharp, melt-bearing velocity transition. These observations are consistent with the southern KP representing the eastern continuation of East Asian lithosphere modified by paleo-Pacific subduction-related thinning and removal since the Early Cretaceous. The continental lithosphere shows systematic regional variations: the SE block exhibits a thicker crust, a shallower LAB, and elevated crustal V_P/V_S relative to the NW block, suggesting localized magmatic and/or thermal overprinting. The adjacent volcanic islands display even more contrasting lithospheric characteristics. Jeju Island preserves inland-like crustal thickness and LAB depths but shows complex crustal V_P/V_S variations and a broad lithosphere–asthenosphere transition, reflecting complex magmatic evolution and distributed melt and/or thermal effects associated with multiple Cenozoic volcanic events. In contrast, Ulleung Island shows exceptionally thin crust, a shallow LAB, and low crustal V_P/V_S , indicating lithospheric modification associated with Miocene backarc opening and post-opening magmatism. Taken together, our findings highlight a tectonically and magmatically modified lithosphere at the transition between the East Asian cratonic blocks and the western Pacific subduction system, linking the present-day structure of the southern KP and adjacent volcanic islands to the long-term evolution of the eastern Eurasian margin.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

Teleseismic P and S waveform data recorded by the KS network are available through the National Earthquake Comprehensive Information System (NECIS; <https://necis.kma.go.kr>). The data from the KG network are accessible through KIGAM Quake (<https://data.kigam.re.kr/quake/data/event-data>), the Big Data Open Platform operated by the Korea Institute of Geoscience and Mineral Resources (KIGAM). Receiver functions can be accessed at the repository (M. Kim, Kang, & Kim, 2026). Data preprocessing was conducted using ObsPy version

1.3.0 (Beyreuther et al., 2010). We used SeisPy version 1.3.4 (Xu & He, 2023) for calculating and processing receiver functions. The thermodynamic modeling was conducted using pHMELTS (Asimow et al., 2004; Ghiorso et al., 2002), implemented in the alphaMELTS v.1.9 package (Smith & Asimow, 2005). Maps were created through PyGMT version 0.14.0 (Tian et al., 2024) based on the Generic Mapping Tool version 6 (Wessel et al., 2019). Synthetic receiver functions were generated using Raysum version 1.2 (Frederiksen & Bostock, 2000) for the 1D velocity model. The lithology information of the southern Korean Peninsula was derived from the 1:1,000,000-scale Geological Map of Korea (Kee et al., 2019), provided by the Korea Institute of Geoscience and Mineral Resources (KIGAM).

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